Jimmy Huang

Senior Backend Engineer & Cloud Architect

Taipei, Taiwan   skyrocketqy81@gmail.com   adminx-ebon.vercel.app   jimmy-huang-07aa4722a
skyrocket-qy   rivendinner

Experience

Fortune Fantasy Global Tech .Ltd, Backend Engineer — Taiwan, June 2025 – present, 11 months
- Developing various modern slot games
- Administrator backend management
- Explore OpenClaw usage with business

Fontech, Senior Backend Engineer — Remote, Nov 2024 – May 2025, 7 months

A real-time, high-throughput, multi-tenant SaaS alarm management service used by various U.S. corporations.
- Built from the ground up
- Spearheaded a team of 3 backend engineers, implementing robust library with clean architecture to improve team's 50% delivery velocity
- Architected critical system components focusing on horizontal scalability
- Integrated various AWS infrastructure (Cognito, EC2, KMS, RDS…etc) to reduce operational overhead
- Developed advanced SQL query optimizations that reduced complex query times from 60s to 0.1s

AMI, BMC Firmware Engineer II — Taiwan, Oct 2024 – Mar 2025, 6 months
- Independently operates in BMC
- Proficient in D-Bus usage
- Experienced and skilled in developing with Redfish APIs
- Implemented optimization library for REP architecture, saving up to 50% time and code to do parse and validate
- Resolved multiple defects across different level modules
- Proactively proposes solutions to work-related issues and shares them with teams
- Collaborate with India and US teams with English communication

Taihe Technology Co., Ltd., Backend Engineer — Taiwan, Mar 2024 – Sept 2024, 7 months
- Implemented new features and maintained existing microservice projects using PHP and Go
- Developed new payment and management system using GoFrame and gRPC
- Quickly familiarized with customized ORM, project architecture, and memSQL
- Assisted juniors in resolving backend and frontend issues

CIeNET Technologies, Software Engineer — On-site at Google's Camera System Team, Sept 2023 – Mar 2024, 7 months
- Responsible for testing Pixel cameras, also developing scripts (Python, Shell) to increase automation test coverage from 85% to 95%
- Analyzed test data for potential root causes
- Maintaining the automated testing pipeline

Ubitus, Backend Engineer

Taiwan
June 2021 – Aug 2023
2 years 3 months

- Development of programs based on microservices and cloud infrastructure (Kubernetes, AWS) using Go, Python, and Node.js.
- Deployment and scripting for setting up cloud services and various services using Docker, Ansible, and Helm.
- Development and maintenance of an existing in-game content management system using Node.js, Go, and Python.
- Establishing a real-time Twitter comment monitoring system for internal use using Python and Prometheus.
- Developing AI-related services and applications like stable-diffusion using Python.
- Built up log collectors and workflow mechanisms using Fluentd and Airflow.
- Writing unit tests, integration tests, and end-to-end tests using K6.

Chung Yuan Christian University, Teaching Assistant

Taiwan
Sept 2021 – June 2022
10 months

- Object Oriented Programming TA
- Introduction to Data Science TA

Education

MS **Chung Yuan Christian University**, Applied Mathematics — Sept 2020 – June 2022

BS **Chang Jung Christian University**, Accounting and Information — Sept 2015 – June 2019

Projects

Alarm system

Associated with Fontech
Nov 2024 – present

The project built on well-known camera corp for alarm management.

- Built the backend infrastructure and db schema to fit business logic.
- Optimized the index creation to ensure low RTs, reducing them from 50s to 3s.
- Built a real-time alarm monitoring API using Go (Gin) and GORM, supporting complex alarm lifecycles and multi-tenant data isolation.
- Integrated AWS Cognito and custom JWT middleware to manage secure user authentication and role-based access control.
- Automated schema migrations and versioning across MySQL using Atlas and GORM.
- Architected a secure Webhook and API Key system to enable third-party integrations and automated reporting.
- Standardized API documentation with Swagger (swag) and implemented structured logging using Zerolog for rapid debugging.
- Containerized the application with Docker and automated deployment workflows using Makefiles and Shell scripts.

AuthZ

July 2025 – Aug 2025

- Architected a real-time, Zanzibar-inspired ReBAC engine supporting complex, recursive permission schemas (RBAC/ABAC), achieving 10k+ RPS per service with sub-1ms median latency.
- Engineered a memory-first, sharded graph architecture that eliminated lock contention and facilitated efficient tree-based relationship exploration for authorization tuple operations.
- Optimized database performance by implementing Redis-backed Cuckoo Filters for probabilistic existence checks, preventing redundant writes and slashing expensive duplicate lookups.
- Built an event-driven state synchronization pipeline utilizing Kafka and Debezium (CDC) to stream tuple updates, ensuring real-time consistency across a distributed cluster.
- Integrated comprehensive observability with OpenTelemetry (OTel) for distributed tracing and metrics, streamlining troubleshooting in high-throughput production environments.

RAG July 2025 – Aug 2025

This project implements a RAG (Retrieval-Augmented Generation) system to accelerate software development tasks like architecture comprehension and debugging. It leverages a large language model (LLM) to understand and analyze a codebase.

- Used LangChain to generate vector embeddings from the source code.
- Stored embeddings in a PostgreSQL database using the PGVector extension for efficient similarity searches.
- A self-hosted Llama model serves as the core LLM for generation and reasoning.
- Retrieves relevant code snippets from the vector store and provides them as context to the LLM upon query.

The application of tree-based model for well interpretation strategy Associated with Chung Yuan Christian University, Jan 2022 – June 2023

- Engineered an automated data extraction pipeline using a web crawler to scrape Grandmaster-level replays, processing game screens every two seconds to create a continuous dataset of 3,309 matches.
- Developed a computer vision module utilizing Hough Circles and custom RGB frame filters to detect and extract dynamic in-game features.
- Scaled the image dataset for model training by applying extensive data augmentation techniques, including background splitting and randomization, to generate over 52,500 training samples.
- Trained and evaluated multiple machine learning models—including CNNs, XGBoost, Random Forest, and GradientBoost—using 8-fold cross-validation to classify game states.
- Designed an interpretable decision-tree-based recommendation engine that analyzed real-time feature importance, successfully identifying strategies that increased simulated win rates by 18% to 34%.

Vistrace

- Engineered a high-performance ARPG engine in Rust/Bevy with strict DOD and 100% ECS compliance, optimizing the WASM pipeline for sub-second web deployment.
- Architected an AI-driven Agentic Workflow utilizing 10+ specialized AI sub-agents to autonomously execute TDD logic, verify builds, and manage cross-module dependencies, reducing development cycles by 30%.
- Developed 12+ custom Rust CLI tools and static analysis guardrails that eradicated circular dependencies and automated 80% of the game's content ingestion pipeline.
- Automated end-to-end testing and documentation by combining AI scenario generation with Playwright/Rust, achieving 95%+ test coverage on core systems, cutting manual QA time by 60%, and maintaining 100% automated documentation accuracy.
- Integrated generative AI pipelines to accelerate asset creation, deploying 30+ production-ready UI and background assets and reducing deployment time from days to minutes.

Licenses and Certifications

AWS Certified Solutions Architect – Associate Amazon Web Services (AWS)

Gemini certified student Google

Skills

Languages: Go, Python, C, Rust, JavaScript (Node.js/React.js/Next.js), Lua, Shell

Infrastructure & Cloud: AWS, Kubernetes, Docker, Helm, Ansible, CI/CD

AI/ML & Data Science: LLM, RAG, TensorFlow, CNN, Decision Trees, Web Crawling

Monitoring & Observability: OpenTelemetry, Prometheus, Grafana, Loki, Fluentd, D-Bus, K6 (Testing)

Operating Systems & Tools: Linux, Git, Makefile, D-Bus

Languages

Mandarin (Native)

English (Intermediate)